Analysis of College Students' Mental Health Based on Machine Learning Algorithm

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***Abstract*—College students are the hope and future of national development, and all walks of life pay more attention to college students. However, the immaturity of the mind, changes in the external environment, employment pressure, and study pressure have caused different types of psychological cognitive disorders among contemporary college students, resulting in deficiencies in emotional regulation, interpersonal communication, and self-character, forming serious mental health problems. Based on machine learning algorithms, this paper analyzes the mental health of college students, obtains the law of college students' mental health data analysis, and establishes a classification model according to the law, so as to quickly screen college students with sub-mental health, adopt psychological counseling to communicate with students with abnormal mental health, and formulate targeted treatment plans for students who have shown obvious mental health problems, help college students solve mental health problems, and cultivate high-quality, mentality and all-round development of psychological data , contribute to the development of the motherland.**

Keywords—machine learning algorithms, mental health; health conditions; Psychological

I. Introduction

As an important branch [1], has gradually entered people's lives. Machine learning is an algorithm that automatically improves performance by letting computers find patterns and patterns in data, and its applications cover various fields [2], including healthcare, finance, transportation, etc. Beyond these practical applications, however, machine learning also has potential implications for people's mental health. This article will conduct research and discussion on the impact of machine learning on the mental health of college students.

A. Definition and development of machine learning

Machine learning refers to the process by which a computer program automatically improves algorithms and improves the accuracy of predictions through learning and training on data. Machine learning techniques can be divided into three types: supervised learning [3], unsupervised learning, and reinforcement learning. Supervised learning is the most common type, the core of which is to predict the labeling of unknown data by analyzing and learning from data with known labels [4]; Unsupervised learning is through the analysis and learning of a large amount of unlabeled data [5], from which the underlying structure and law are found; Reinforcement learning refers to interacting with the environment to allow computers to automatically learn and optimize decision-making processes.

With the continuous development of technologies such as deep learning, the application of machine learning in various fields is becoming more and more extensive. At present, machine learning has become one of the most promising and promising areas of artificial intelligence technology [6].

B. The relationship between machine learning and mental health

Although machine learning is becoming more widely used in various fields, it is still controversial whether it will have an impact on people's mental health [7]. The following explores the impact of machine learning on college students' mental health from the following two aspects:

1) The impact of machine learning on psychotherapy: Traditional psychotherapy often relies on the experience and judgment of human physicians [8]. However, due to problems such as cognitive bias and misdiagnosis among human doctors, the results of traditional psychotherapy are not very satisfactory. With the continuous development of machine learning technology, some new psychotherapeutic methods based on machine learning have also begun to emerge. For example, natural language processing technology and machine learning algorithms are used to analyze patients' psychological problems and symptoms to provide more accurate and personalized treatment plans. The application of these new technologies can effectively alleviate the shortcomings in traditional psychotherapy, improve the treatment effect and patient satisfaction [9].

2) The impact of machine learning on psychological stress: With the continuous change and development of society, the psychological pressure faced by modern people is also increasing. For the group of college students, they are in a critical period of growth [10], facing problems such as academic pressure, employment pressure, and interpersonal pressure. These problems not only affect their physical health, but also negatively affect their mental health. If machine learning is applied to the field of mental health, it may provide some new ideas for solving these problems [11]. For example, machine learning algorithms are used to analyze a large number of psychological literature and research data to discover factors and laws related to psychological stress; Or use machine learning technology to develop some psychological stress monitoring and management tools, timely warning and relief of individual stress levels and so on. The application of these new technologies can help college students better manage their emotions and stress to maintain a good state of mental health [12].

This paper mainly discusses and studies the impact of machine learning on the mental health of college students. The results show that machine learning technology can improve traditional psychotherapy to a certain extent, and provide new ideas and methods for solving problems such as psychological stress of college students [13]. However, we should also see that machine learning technology itself also has certain limitations, such as data privacy protection, algorithm fairness

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and other issues need to be further studied and solved. In the future, we need to continue to explore and innovate to better play the application value of machine learning technology in the field of mental health.

II. Related Concepts

A. Mathematical description based on machine learning algorithms

Data mining technology can extract valuable information from a large number of students' information, and then evaluate students' mental health status. For example, by analyzing students' social network behavior, online search records and other data, we can evaluate their psychological state. is shown in Equation (1).

$$G = \frac{1}{\frac{1}{C_e s} + L_k s + R_k + T} \tag{1}$$

Health is a complex process, and it is also a continuous data monitoring is shown in Equation (2).

$$\hat{x} = (A^T A)^{-1} A^T b \tag{2}$$

Combined with traditional psychological evaluation methods, such as SCL-90, we can understand students' psychological status more comprehensively. This method of psychological assessment based on data mining can not only improve the efficiency of assessment, but also provide a basis for the follow-up psychological intervention as shown in Equation (3).

$$y[n] = \sum_{m=-\infty}^{+\infty} x[m]\, h[n-m] \tag{3}$$

B. Choice of communication skills program

Natural language processing technology can be used to intelligently analyze the language of college students, so as to find out their potential psychological problems as shown in Equation (4).

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \tag{4}$$

Based on hypotheses I and II, the mental health synthesis function is obtained, and the result is shown in Equation (5).

$$y[n] + RMSE \leq \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{5}$$

By analyzing students' text expression, pronunciation and intonation, we can judge their emotional state and personality characteristics. are shown in Equation (6).

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\hat{y}_i - y_i}{y_i}\right| \tag{6}$$

C. Screening program for mental health abnormalities

Combined with this technology, psychological counseling teachers can communicate with students more pertinently and provide personalized psychological counseling suggestions. This method can greatly improve the efficiency and effect of psychological counseling. are shown in Equation (7).

$$U_{1h} = \frac{I_{1h}}{2\pi h f_0 C_1} \angle\left[\varphi_{1h} - \frac{\pi}{2}\right] \tag{7}$$

Mental health should also be analyzed and judged between different data.shown in Equation (8).

$$\theta_{sh} = -\,arctan\left(\frac{U_{1h}\cos\varphi_{1h} + U_{2h}\cos\varphi_{2h}}{U_{1h}\sin\varphi_{1h} + U_{2h}\sin\varphi_{2h}}\right) \tag{8}$$

Machine learning technology can analyze and learn the psychological data of college students, so as to predict their future psychological state as shown in the formula $j\omega$ (9).

$$U' = \frac{C_1}{C_1 + C_2} U_1 - I\left(j\omega L_\Sigma \quad - \frac{1}{j\omega(C_1 + C_2)}\right) \tag{9}$$

By establishing a prediction model, students with potential psychological problems can be warned and relevant personnel can be reminded to intervene in time. This prediction model is helpful to find out students' psychological problems in advance and provide them with timely help and support.

III. Strategies For Optimizing The Mental Health Of College Students

Health should first select the corresponding samples, judge the corresponding psychological content and psychological data, then form a relatively comprehensive data structure, analyze and fuse the sample structure and content to realize selective analysis of samples, and then screen the necessary contents and conditions in the samples.To form more effective sample data and result in sexual health, we should judge and screen according to actual standards, and find out that there are significant differences in data formation and comprehensive samples.

IV. Practical Cases Of Mental Health In College Students

A. Mental Health Exception Screening Fact Sheet

It is necessary to observe the behavior, speech and emotion of college students to understand their daily life, study and interpersonal communication. In addition, students' personal information, family background, study pressure and other data can be collected through questionnaires and interviews. is shown in Table I.

TABLE I. Screening Requirements For College Students For Mental Health Abnormalities

Influencing factors	Psychological qualities	Innovative effect	Communication skills
Communication skills	66.7893	40.7943	42.1811
Stressful events	67.4048	42.8182	43.4795
Academic performance	67.3162	43.1974	42.9932
Single parent families	66.8568	42.7591	38.5975
Emotional experiences	66.5235	41.9364	41.0369

The screening process for mental health abnormalities in Table I. is shown in Figure I.

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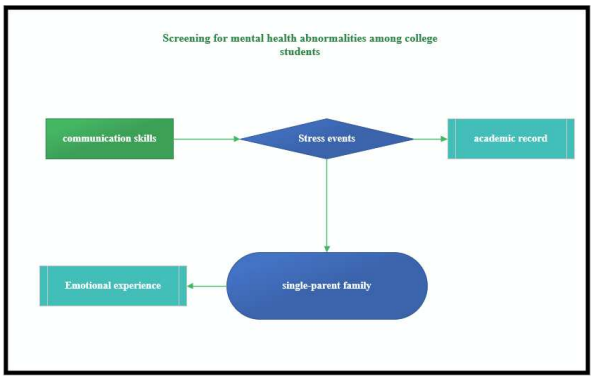

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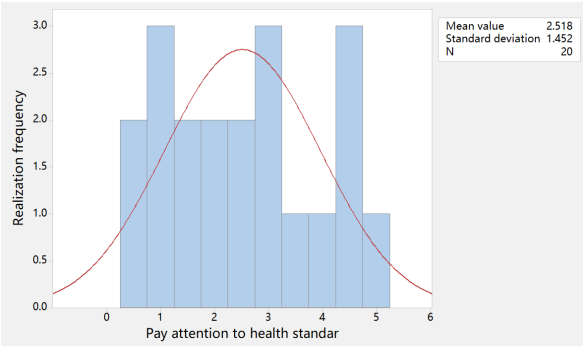

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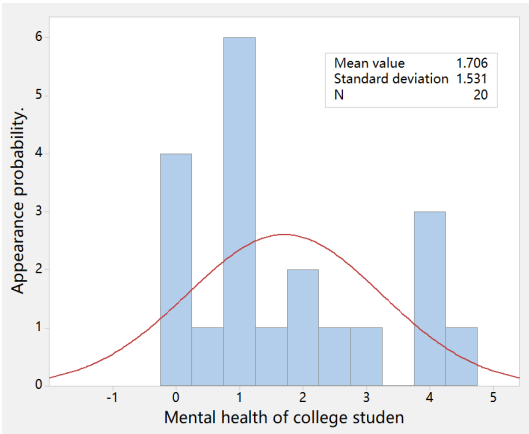

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Fig. 1. Analysis of the mental health of college students. (a) Process, (b) Data processing, (c) Data change process

According to the observation and collected information, the college students are screened preliminarily, and the students who may have psychological problems are found. This process can be assisted by setting some key indicators, such as abnormal mood, interpersonal tension, declining academic performance, etc.

B. *College student mental health*

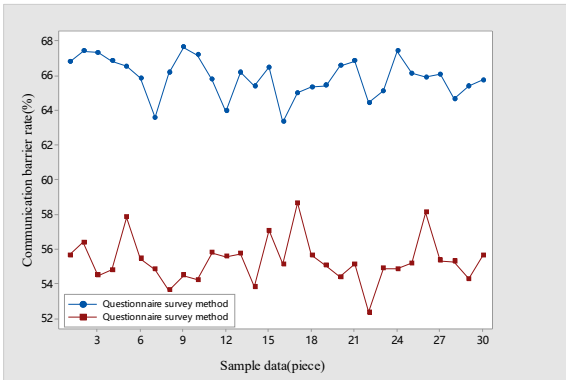
For the students who have psychological problems, they need to be evaluated more professionally and deeply. This usually includes using psychological assessment tools (such as SCL-90, SAS, SDS, etc.) to measure, and inviting psychological counselors or experts to conduct interview evaluation is shown in Table II shown.

TABLE II. THE OVERALL PICTURE OF THE COMMUNICATION COMPETENCY PROGRAM

Influencing factors	Communication barrier rate	Rates of psychological disorders
Communication skills	66.1243	52.3319
Stressful events	65.9140	54.8517
Academic performance	66.0913	54.8377
Single parent families	64.6541	55.1840
Emotional experiences	65.3915	58.1051

C. *Communication ability and stability of screening for mental health abnormalities*

It can help to judge students' mental health status more accurately and identify specific types of mental health problems, such as anxiety, depression, interpersonal distress and so on. is shown in Figure II.



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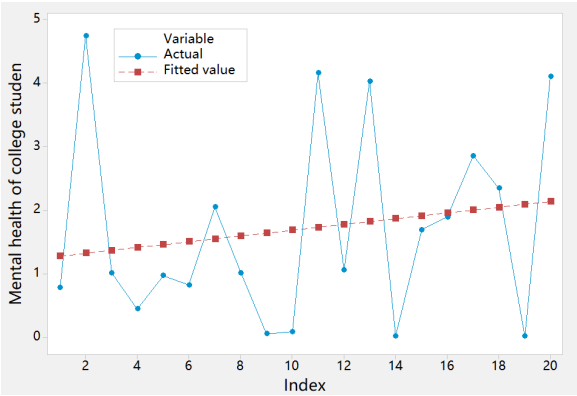

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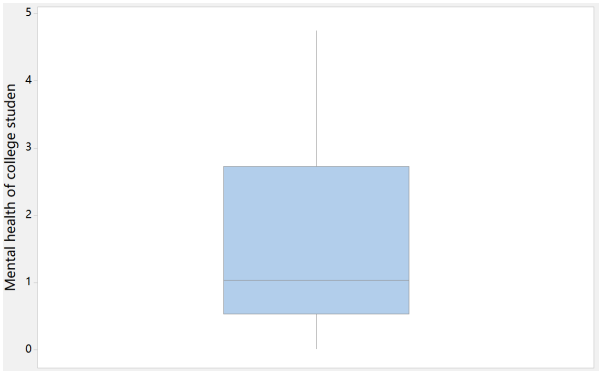

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(c)

Fig. 2. The ability to communicate with different algorithms. (a) Multi-method data comparison, (b) The results of your data, (c) Distribution of national results

Feedback the evaluation results to students in an appropriate way to help them understand their psychological status is shown in Table III.

TABLE III. COMPARISON OF THE ACCURACY OF SCREENING FOR MENTAL HEALTH ABNORMALITIES BY DIFFERENT METHODS

algorithm	Communication skills	Affordability	error
Based on machine learning algorithms	90.8696	92.6958	91.1076
Questionnaire method	55.7673	55.7120	57.0552

For the problem part of the evaluation results, it is necessary to explain and guide in detail, so that students can realize their own problems and provide corresponding suggestions and help Figure III shown.

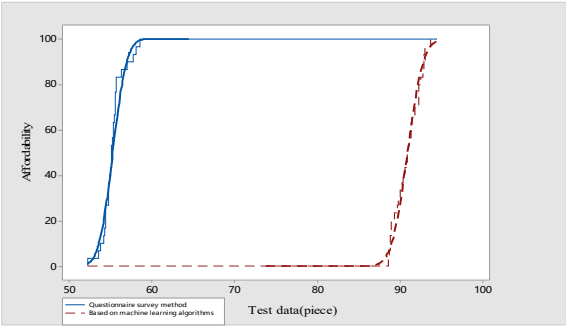

Fig. 3. Communication ability based on machine learning algorithm mental health abnormality screening

According to the evaluation results, make personalized intervention programs for students with psychological problems. This may include psychological counseling, psychotherapy, drug therapy and other ways, aiming at helping students solve psychological problems and improve their mental health level

V. CONCLUSION

With the development of intelligent technology, its application in the field of college students' mental health is more and more extensive. Through data mining, natural language processing and machine learning, we can understand the psychological status of college students more comprehensively and provide them with personalized psychological counseling and intervention. However, the application of intelligent technology also faces some challenges, such as data privacy protection, technical ethics and so on. Therefore, in the future research, we need to further explore how to make rational use of intelligent technology on the basis of protecting students' privacy, and provide better support and services for college students' mental health. At the same time, we also need to pay attention to the limitations and potential risks of technology to ensure the rational application and development of intelligent technology in the field of mental health

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